Milky Way Neutral Hydrogen

Background Drift Correction

Background Drift Stabilization is extremely important to being able to
properly present and visualize hours of hydrogen-line data.
This correction is implemented in HLine3D.
It maintains calibrated amplitudes between Cold_Sky and H1 amplitude,
and normalizes the data at a user specified cold sky region not containing hydrogen spectra.

In addition to providing excellent 'webpage' graphics, these signal processing techniques
create repeatable, calibrated scaling of the data that can be used to quantitatively evaluate
changes in system configurations and hardware designs,
and in comparison of different systems to each other.

https://github.com/AP-HLine-3D/HLine3D
https://github.com/AP-HLine-3D/HLine3D/blob/main/Info_HL3D_Cold_Sky_Drift_Correction_mx%2Bb01.pdf

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Milky Way Neutral Hydrogen

Background Drift Correction

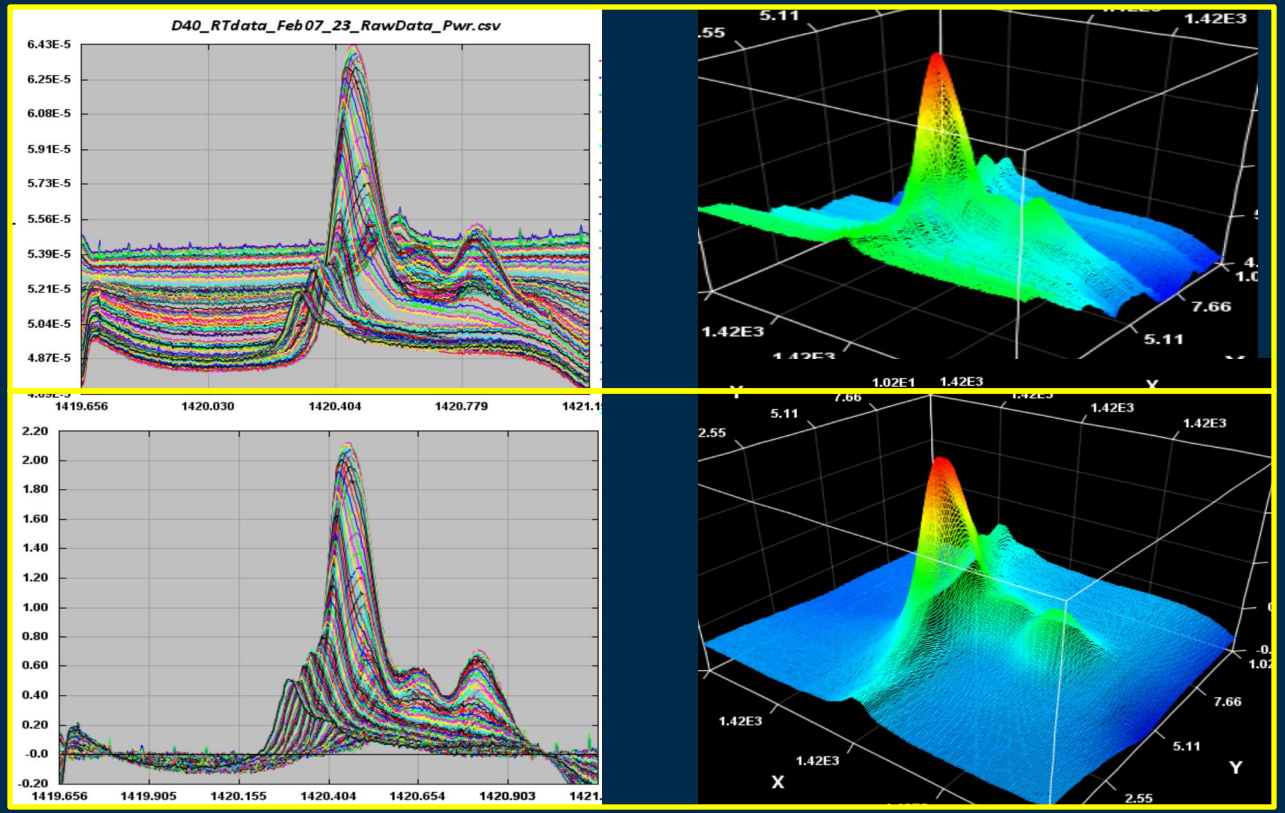
Potential Small Radio Telescope Background Drift/Shift Sources

Ground Noise Spillover
Outside Temperature Variation
Sky Transparency Changes
Sun Blackbody Radiation in Antenna Beam Pattern
Sun reflection off roofs / buildings / trees
Radio Frequency Interference (LAN Blocking)
RF Connection Thermal Expansion Ohmic Changes
Thermal LNA Gain Changes
Thermal SDR Filter Shape Changes
Sunlight Refraction off Clouds

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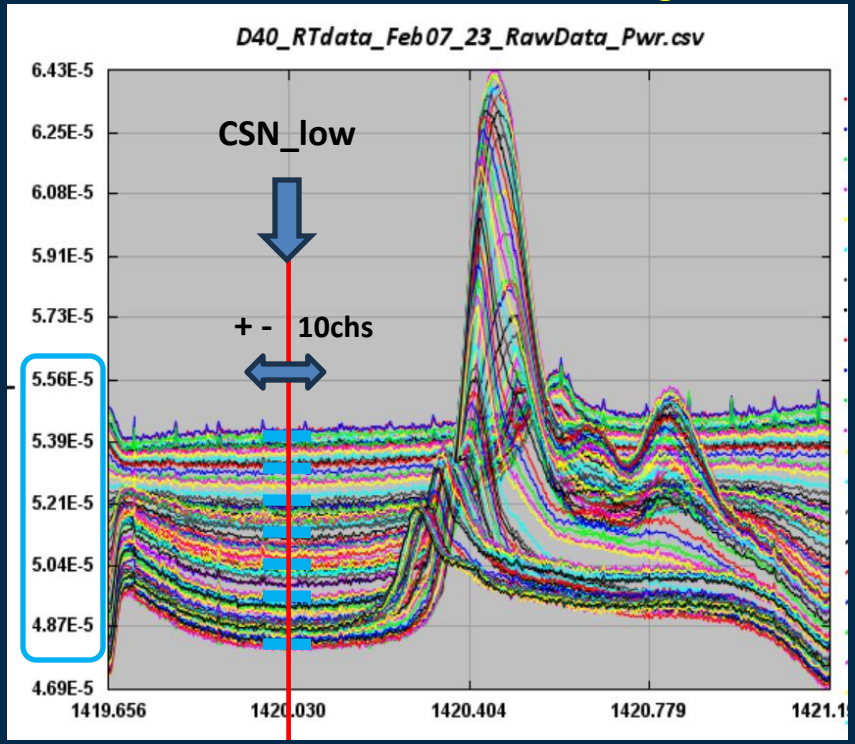
Background Drift Correction

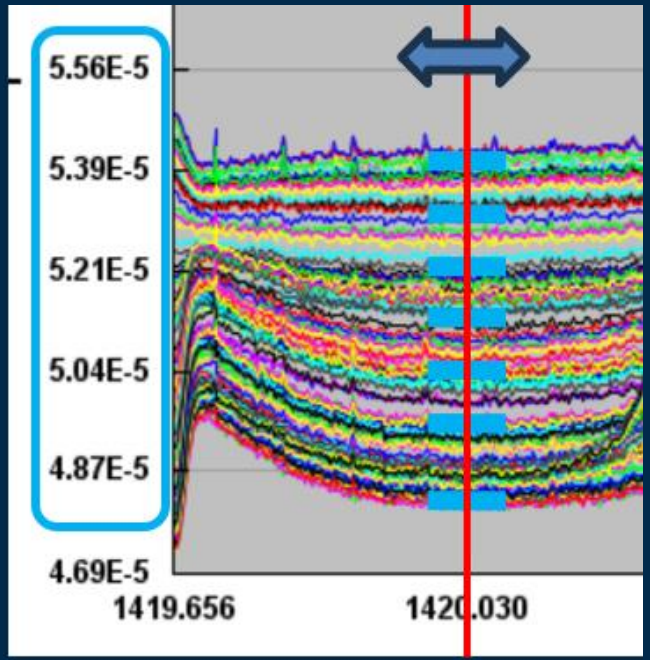


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“ Cold Sky” Background Drift Correction

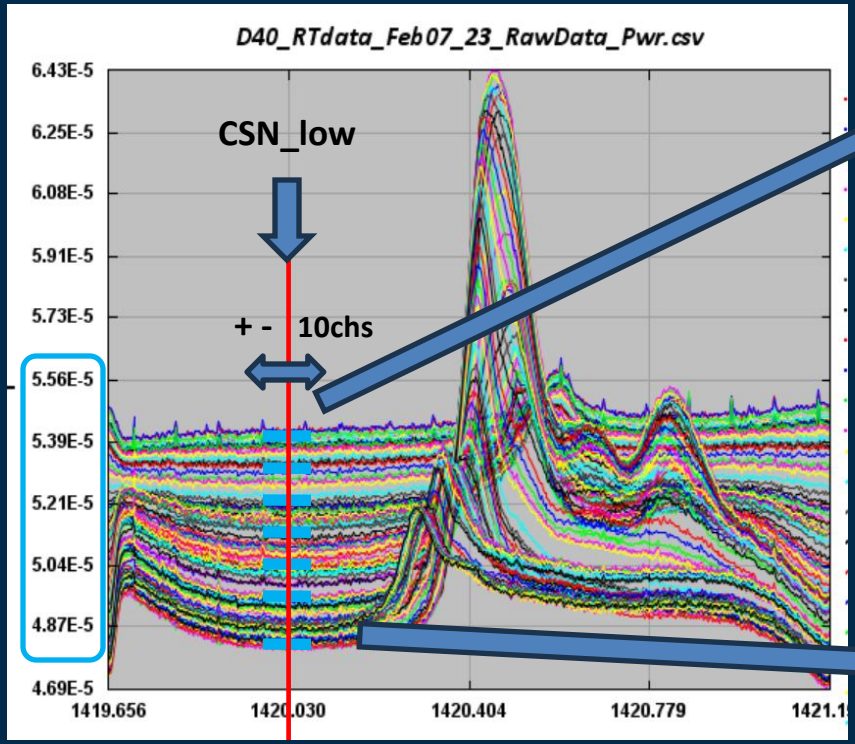
Convert IFavg data to Power (square the amplitudes)
Select a region of Cold Sky &
Calculate the average Power Level over +- 10 channels about that frequency

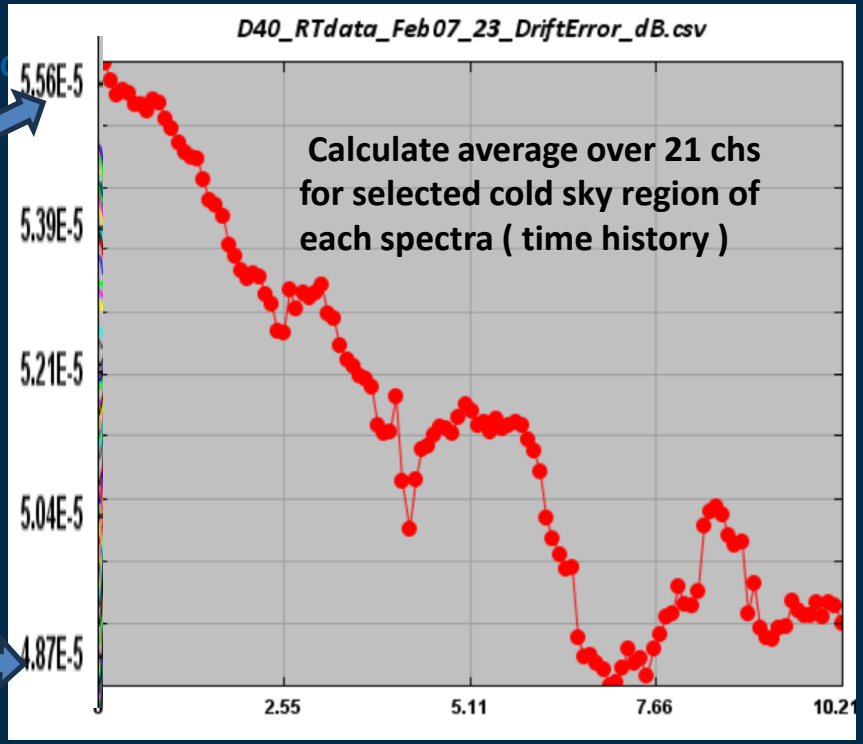




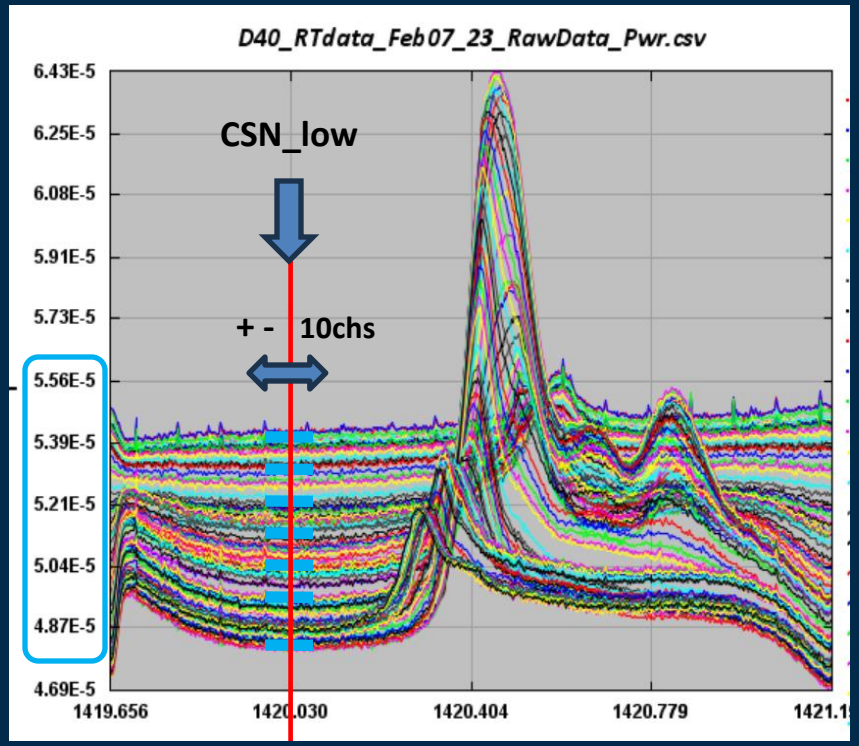
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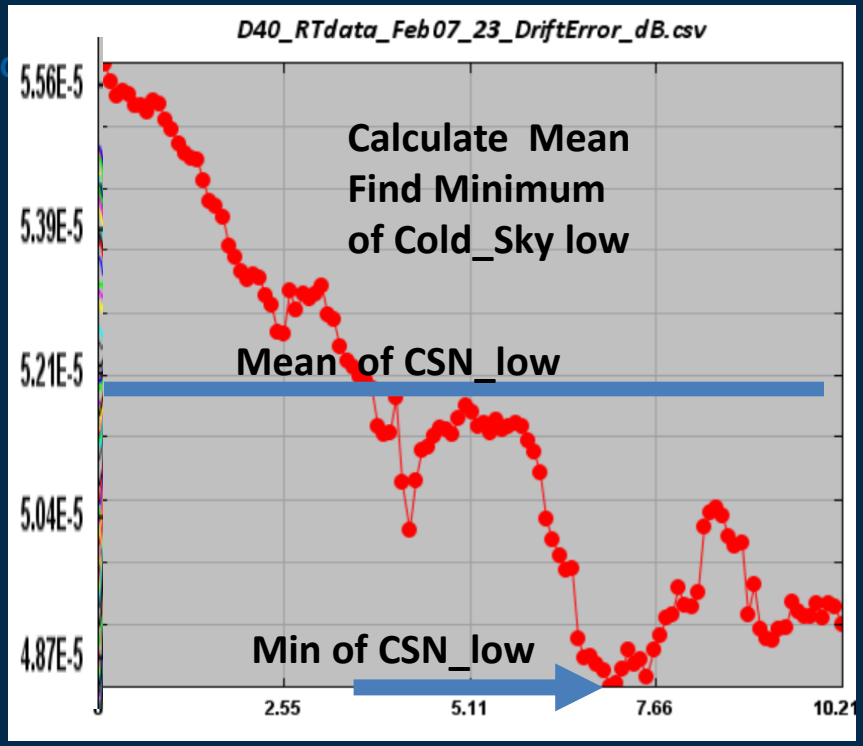
“ Cold Sky” Background Drift Correction





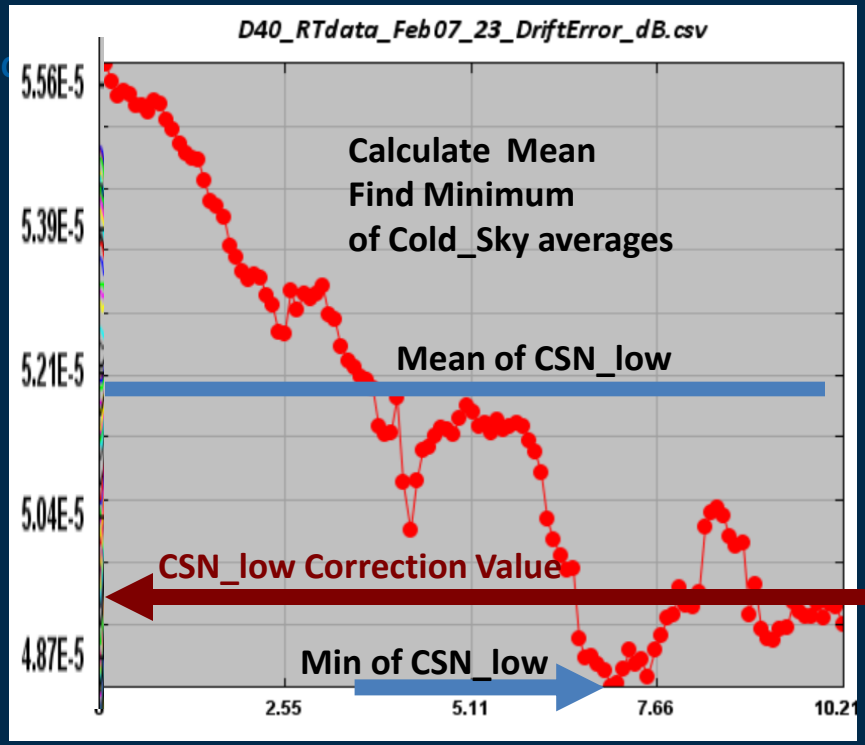
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“ Cold Sky” Background Drift Correction



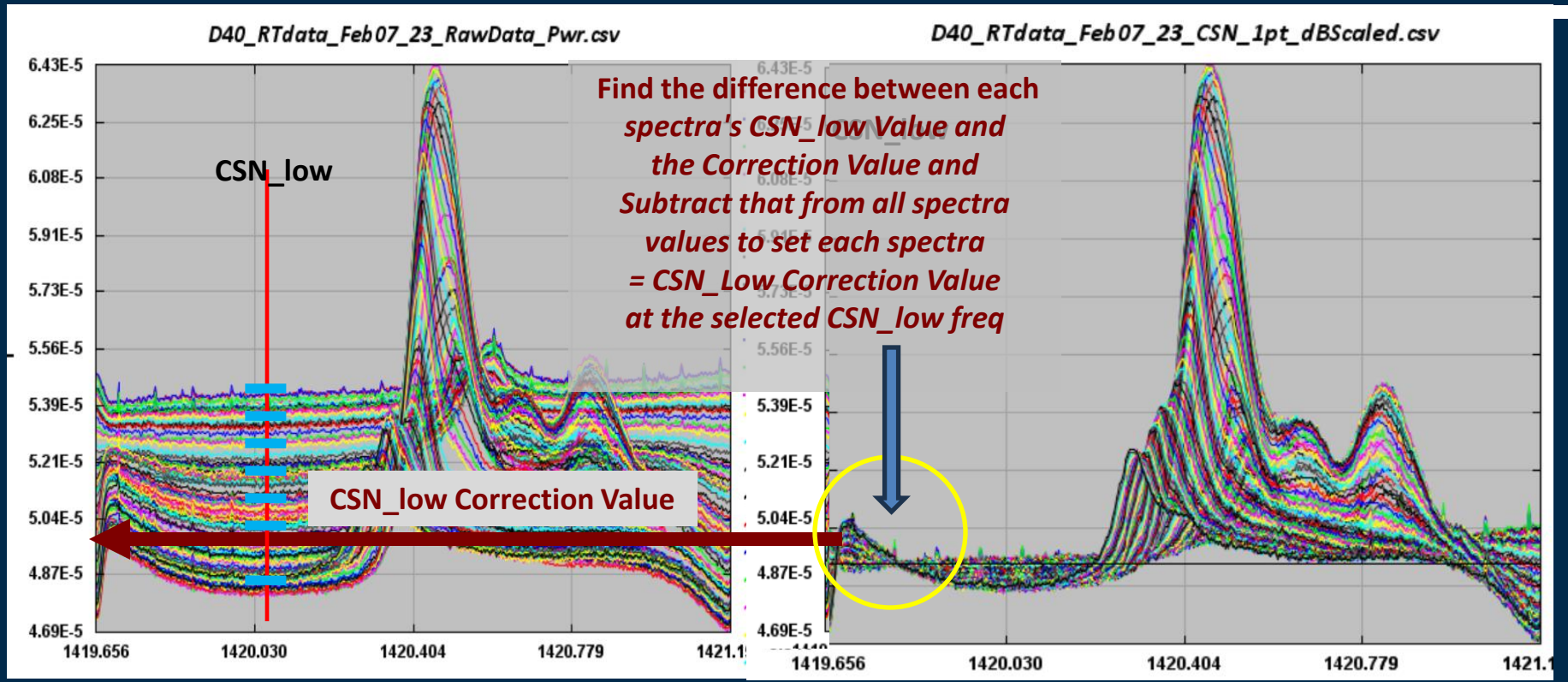
Using a percentage of the Mean in addition to the Minimum provides more stable results in maintaining calibrated *H1 pk over Cold_Sky calculations*

Calculate

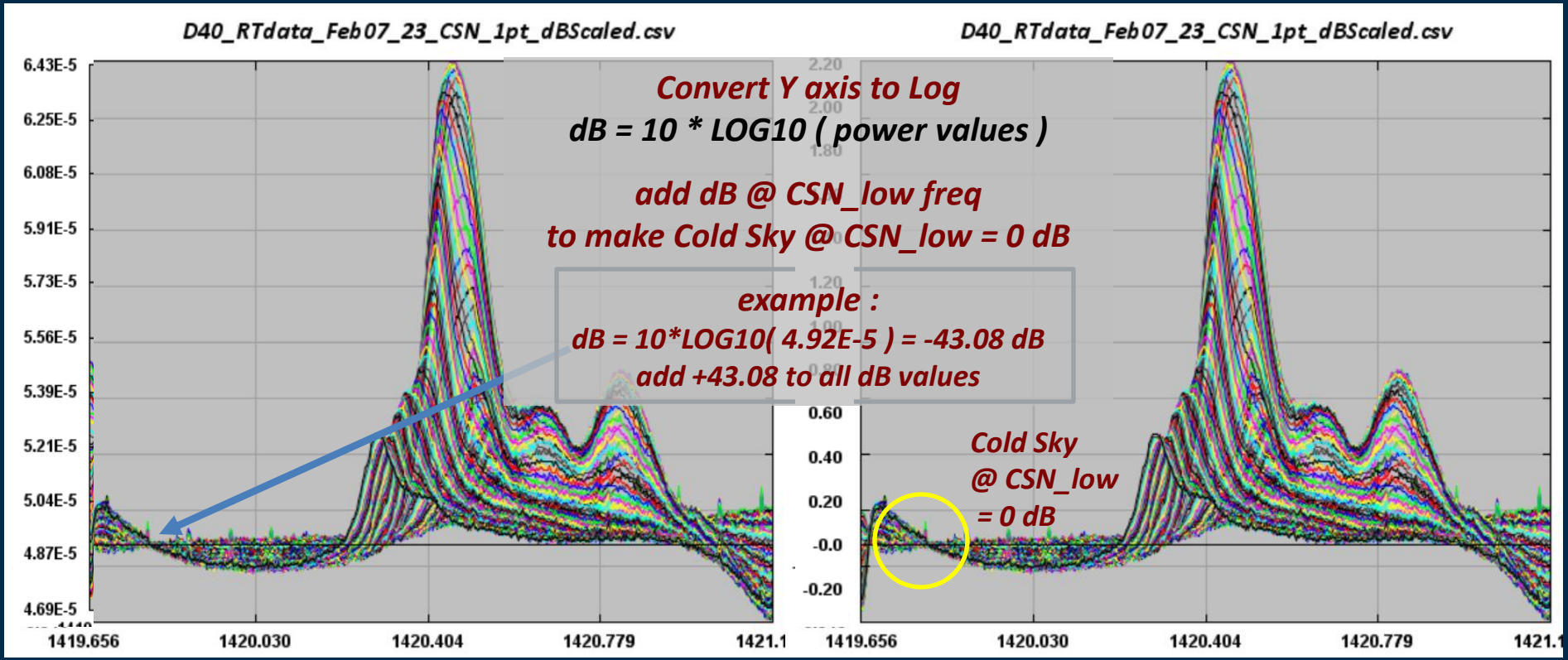
Cold Sky Low Correction Value =

(0.33) * (Mean - Min) + Min

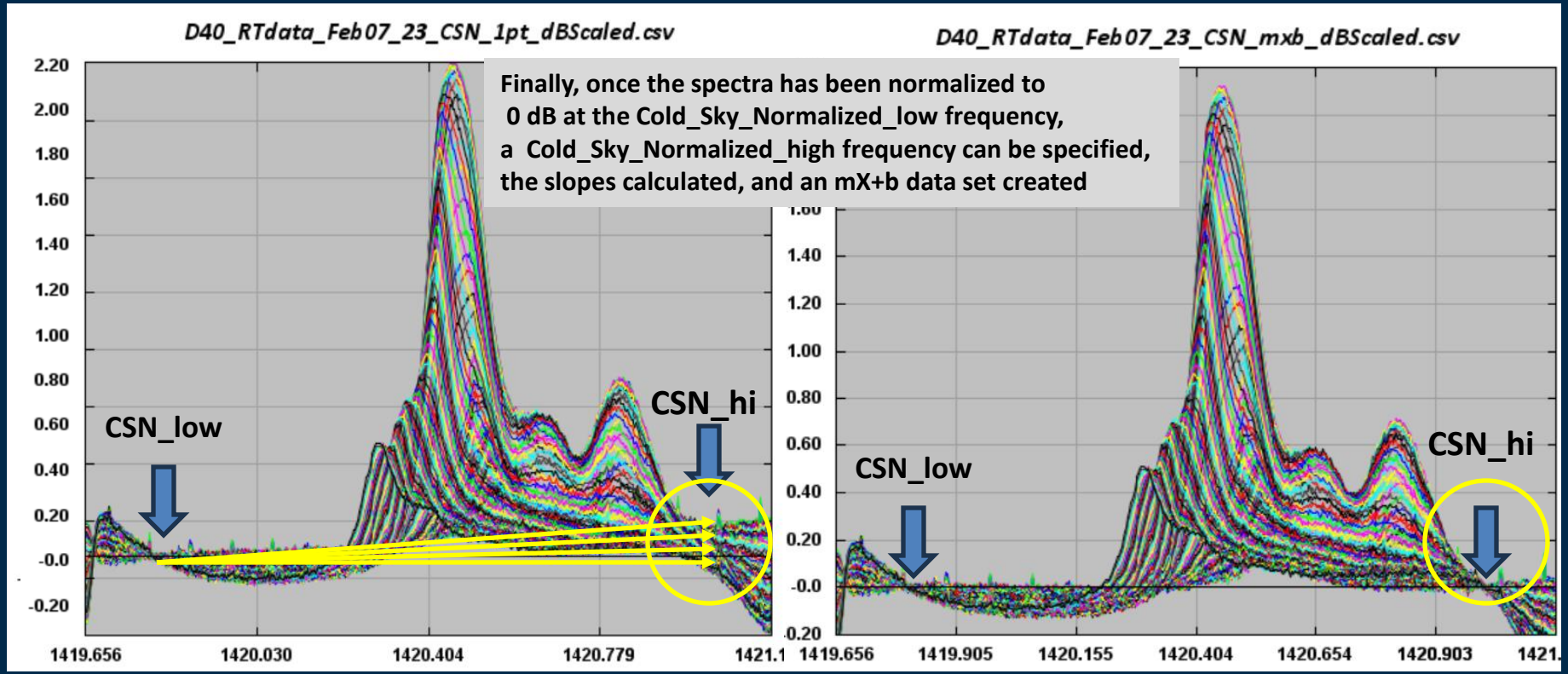
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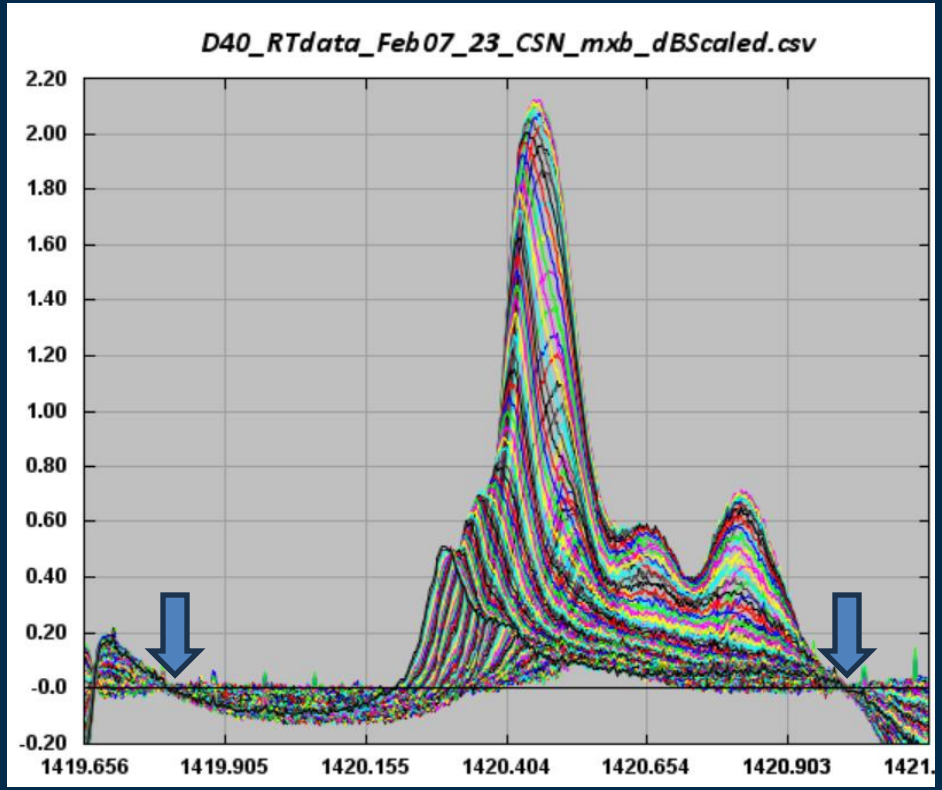
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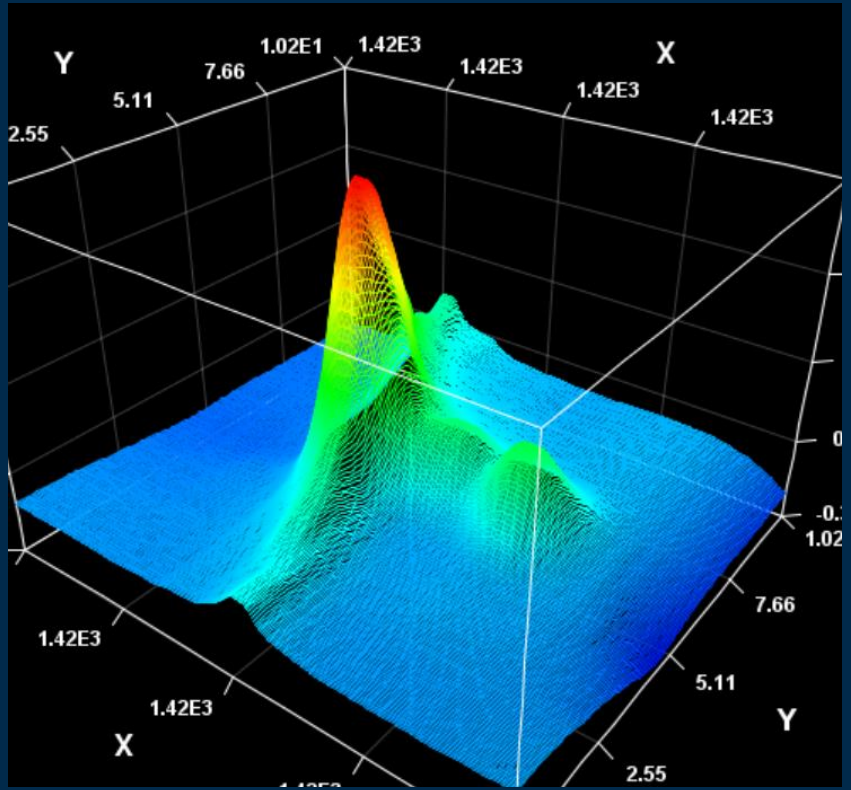


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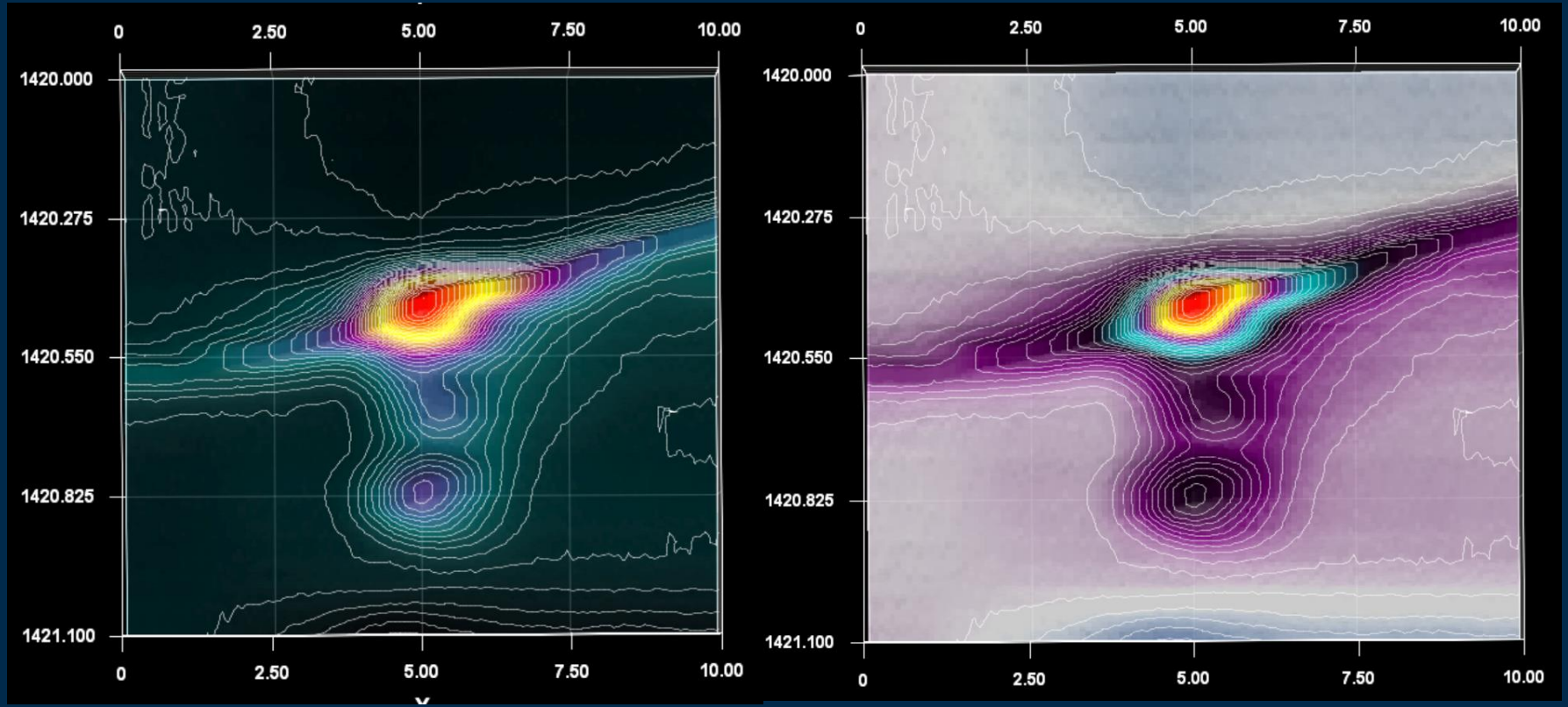
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